

Worksheet: work out True or False before you run it

NAME _____

DATE Mon Sep 28 or Fri Sep 25 _____

CHUNK 4 OF 6

and: both must be True

`and` is True only when both sides are True.

CHUNK 4

```
height = 52
has_ticket = True
if height > 48 and has_ticket:
    print("Board the coaster.")
else:
    print("Step aside.")
```

Each row is one rider. Fill in the blanks. Shaded rows are done for you.

rider	height	has_ticket	height > 48?	height > 48 and has_ticket?	what it prints
1	52	True	True	True	Board the coaster.
2	52	False			
3	48	False			
4	48	True	False	False	Step aside.
5	40	True			
6	60	False			
7	49	False	True	False	Step aside.
8	49	True			

Which riders board the coaster? Why?

Run it

Put rider 2's values in lines 1 and 2, then run. Do the same for rider 8. Do not erase a wrong prediction.

Run	What I predicted	What it printed	Right?
rider 2			
rider 8			

CHUNK 5 OF 6

or: one is enough

`or` is True when either side is True.

CHUNK 5

```
is_staff = False
has_pass = True
if is_staff or has_pass:
    print("Gate opens.")
else:
    print("Gate stays shut.")
```

Each row is one visitor. Fill in the blanks. The shaded row is done for you.

visitor	is_staff	has_pass	is_staff or has_pass?	what it prints
1	False	True	True	Gate opens.
2	True	False		
3	True	True		
4	False	False		

Which visitor is kept out? Why?

Run it

Put visitor 4's values in lines 1 and 2, then run. Do not erase a wrong prediction.

Run	What I predicted	What it printed	Right?
visitor 4			

If you have time

Change **or** to **and** on line 3. Which visitors now get a different line?

A while loop and its stopping condition

`!=` means is not equal to. `""` is empty text.

CHUNK 6

```
answer = ""
while answer != "yes" and answer != "no":
    answer = input("Ready? Type yes or no: ")
print("Recorded:", answer)
```

Someone types maybe, then soon, then yes. Each row is one check of the condition. Fill in the blanks. The shaded row is done for you.

check	answer	answer != "yes"?	answer != "no"?	answer != "yes" and answer != "no"?
1	""	True	True	True
2	"maybe"			
3	"soon"			
4	"yes"			

How many times is the question asked? Which check stops the loop?

Run it

Predict each line, then run it and type maybe, soon, yes. Do not erase a wrong prediction.

Line	What I predicted	What it printed	Right?
line 1			
line 2			
line 3			
line 4			

The same loop with or

Line 2 now says `or` . Someone types maybe, then yes, then no. Trace it on paper only.

CHUNK 6, LINE 2 WITH OR

```
while answer != "yes" or answer != "no":
```

check	answer	answer != "yes"?	answer != "no"?	answer != "yes" or answer != "no"?
1	" "	True	True	True
2	"maybe"			
3	"yes"			
4	"no"			

Does this loop ever stop? Why?